

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ Serial0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _ Yes _

Does R2 show up as an OSPF neighbor on R3? _ No _

What does this information tell you?

R2 is only sends OSPF Hello packets out of its S0/0/0 interface to R1. R3 has no direct adjacency with R2 and must route through R1 to reach R2's LAN.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ Serial0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

R3 outgoing S0/0/1 cost 64 + R2 outgoing G0/0 cost 1 = 64 + 1 = 65.

Is R2 listed as an OSPF neighbor to R3? _ Yes _

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

The default OSPF reference bandwidth is 100 Mb/s, which results in a cost of 1 for links running at 100 Mb/s or faster. Increasing the reference bandwidth allows OSPF to differentiate costs on modern higher-speed links

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: The cost is 782. R1 S0/0/1 was set to bandwidth 128 + R3 G0/0 781+1 = 782

192.168.23.0/30: The cost is 845. R1 S0/0/1 + R3 S0/0/1 781 + 64 = 845

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

ter applying bandwidth 128 to all serial interfaces, each serial link has cost 781. The route traverses two serial links, so $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The direct path via S0/0/1 was manually assigned a cost of 1565, making its total cost 1566 ($1565 + 1$). The alternate path via R2 costs 1563 ($781 + 781 + 1$). OSPF chooses the lower-cost path, so it routes through R2.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

It ensures predictability and stability in the OSPF domain. If the router ID is tied to a physical interface, it can change when that interface goes down; using a static router ID or a loopback interface prevents this.

2. Why is the DR/BDR election process not a concern in this lab?

The serial links are point-to-point networks, and OSPF does not perform DR/BDR elections on point-to-point links.

3. Why would you want to set an OSPF interface to passive?

To prevent OSPF Hello packets and route updates from being sent out of that interface, reducing unnecessary traffic and CPU usage on LANs that do not contain other OSPF routers.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of OSPF areas.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

Routers choose between routing protocols using Administrative Distance. EIGRP is lower than OSPF and RIP, so the EIGRP route is preferred regardless of metric.